Comparison Between Cloud SQL and Cloud Storage for Reading RAW data

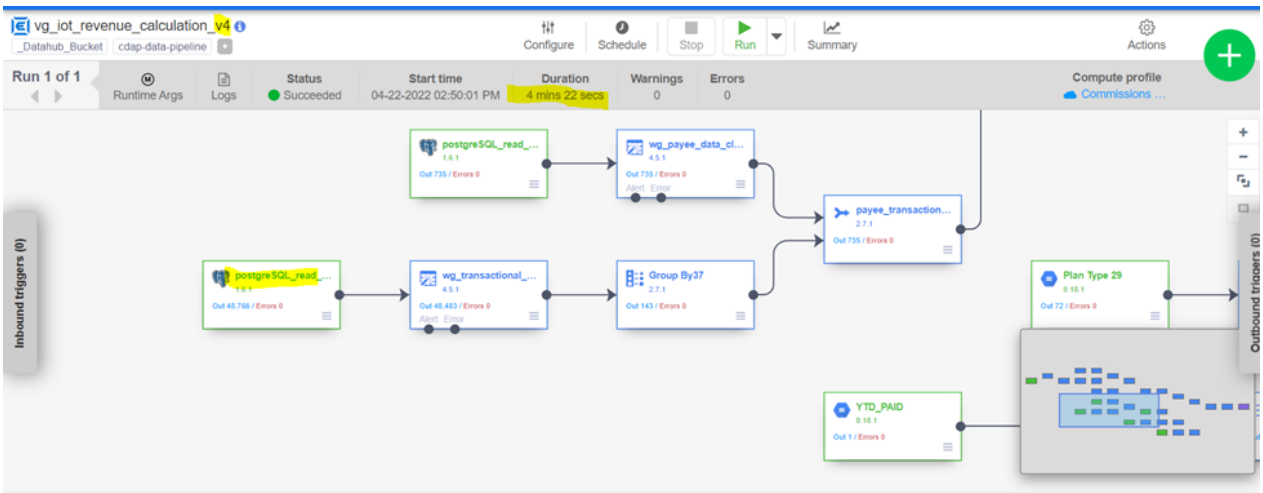
We need to read the revenue transformation pipeline output i.e., raw data in revenue calculation pipeline.

The revenue transformation pipeline output is saved in cloud sql table. Now there are two approaches to consume this data into the calculation pipeline.

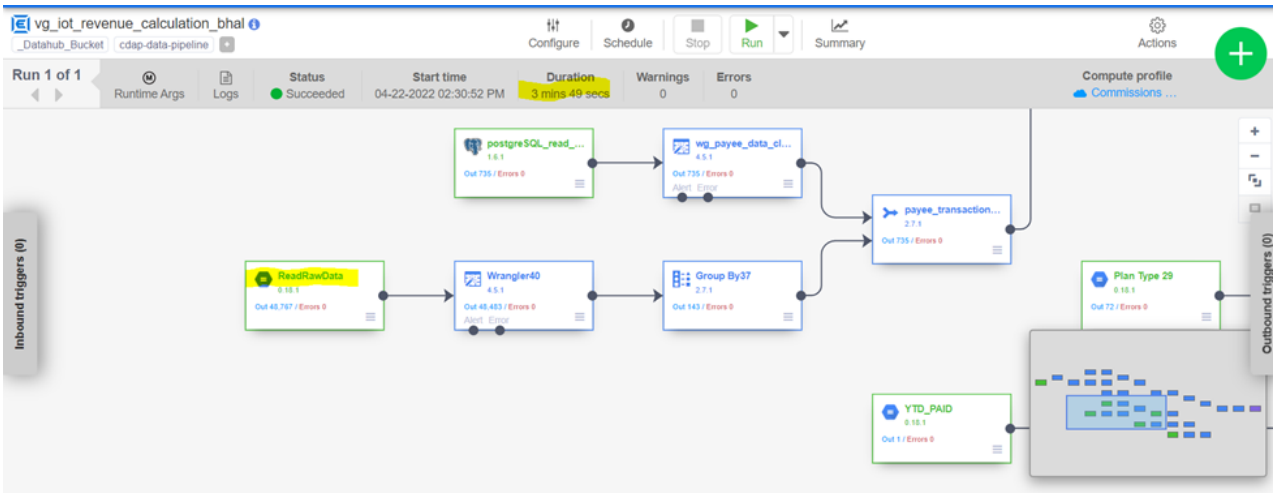
1. Read raw data directly from cloud SQL table using PostgreSQL plugin.
2. Export cloud SQL table into google cloud storage in csv file format and read the csv file in calculation pipeline.

We created 2 Data fusion pipelines for above mentioned approaches and created below analysis.

1] Data fusion pipeline took 4 mins 22 secs to complete the execution for the first approach.



2] Data fusion pipeline took 3 mins 49 secs to complete the execution for the second approach.



There is a difference of **33 seconds** in execution time of both the approaches.

But for second approach we need to work on below 4 additional things which are not required in first approach as we are directly reading data from cloud sql table in the first approach.

1] We need to create a functionality on front end which will provide some notification/ACK file once the user completes exclusion raw data activity on front end.

(**<u>Exclusion raw data activity information:</u>**

Once transformation output is saved in cloud SQL then user will set the column **is_excluded** to true for the transaction records which he/she wants to exclude from

the commission calculation)

2] We need to create an event-based system on back-end part which will get notified using notification which front end team will provide in first point.

3] An event-based system will trigger a dag which will export data from cloud sql table to datahub project gcs bucket.

(Cloud sql instance is in compute project and storage buckets are in datahub project)

4] Also this approach will introduce an additional cost of saving files in GCS

Now there in one more approach which will remove event-based functionality which is as follows.

1] We can have a export task in the revenue calculation pipeline itself.

2] This will first export cloud sql data into gcs and then trigger pipeline for revenue commission calculation.

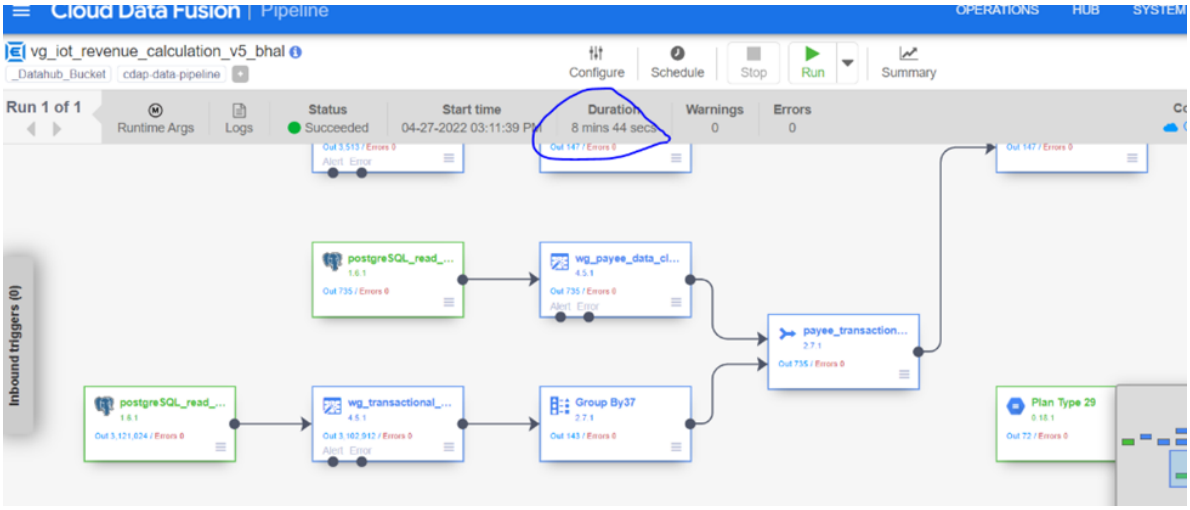
3] This way we don't need to implement event based functionality.

We have performed one more analysis using high volume of data. PFB analyzed things for the same.

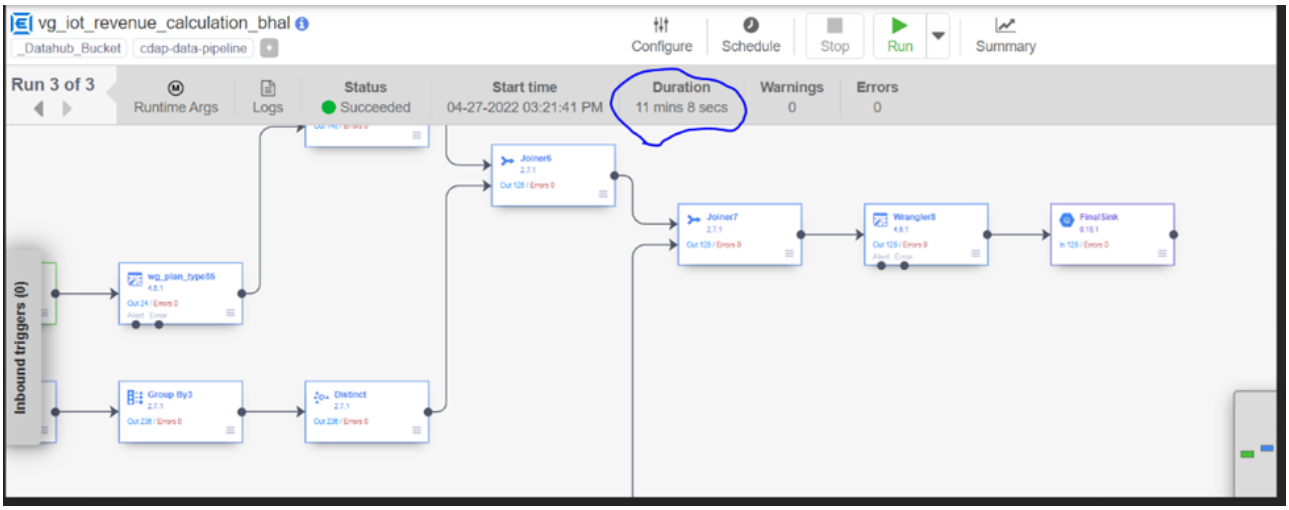
We have run the pipelines with 3121024 records using running cluster and reading from table took 8 mins 44 sec and reading from file took 11 min 8 secs.

PFB screen shot for your reference.

1] Reading from table:



2] Reading from file :



NOTE : In the 1st approach , we have used parallel splits to read the data from the cloud SQL.